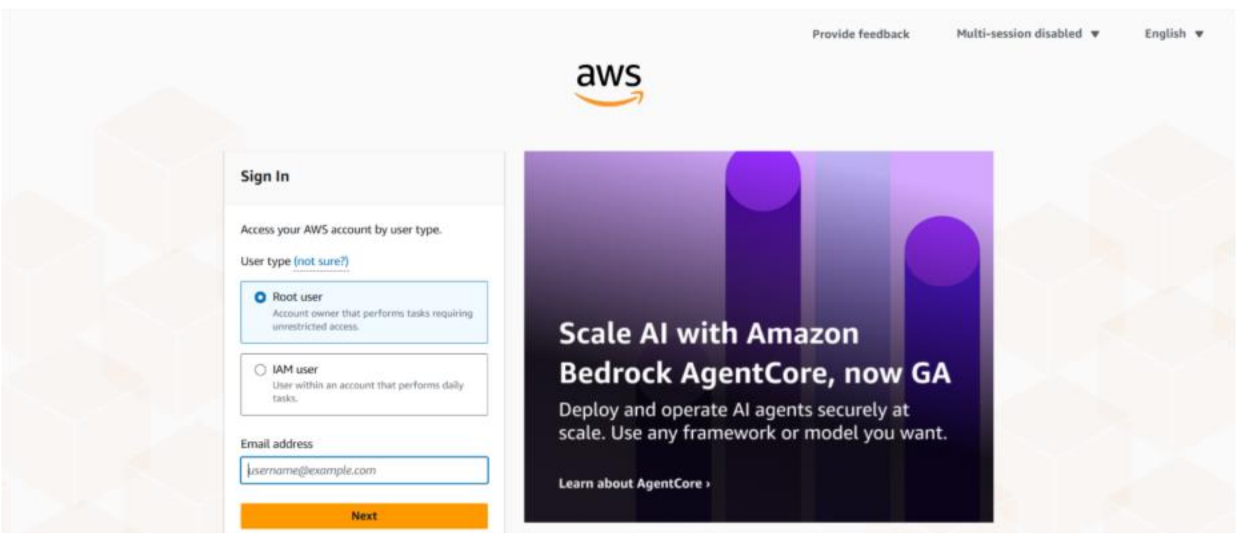
Task-4: Auto Scaling & Load Balancer

1. Create two private EC2 instances in different Availability Zones.
2. Configure Apache (httpd) on the instances.
3. Create an Application Load Balancer (ALB).
4. **Create an Auto Scaling Group and attach it to the Load Balancer.**
5. Verify that:
    a. The website is accessible using the Load Balancer DNS.
    b. The application is still accessible even if one EC2 instance is stopped.

Solution:

1. Login to your AWS account with credentials and required verification.

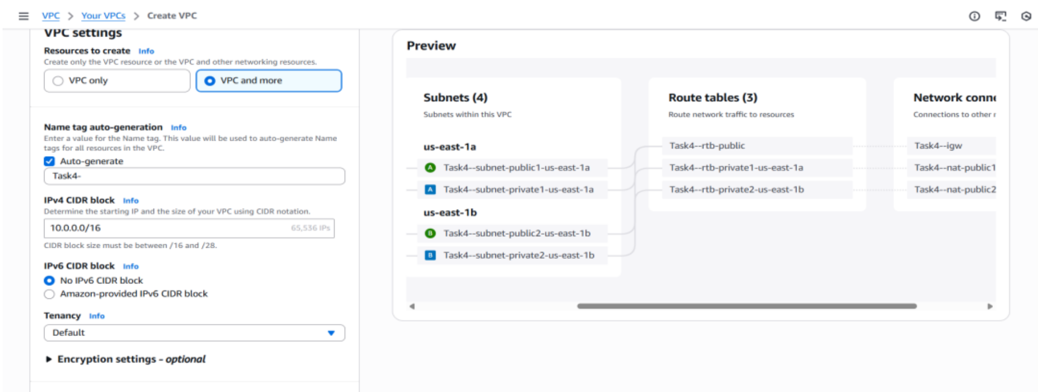
    (https://signin.aws.amazon.com/)
    My account – AWS Management Console- Sign in



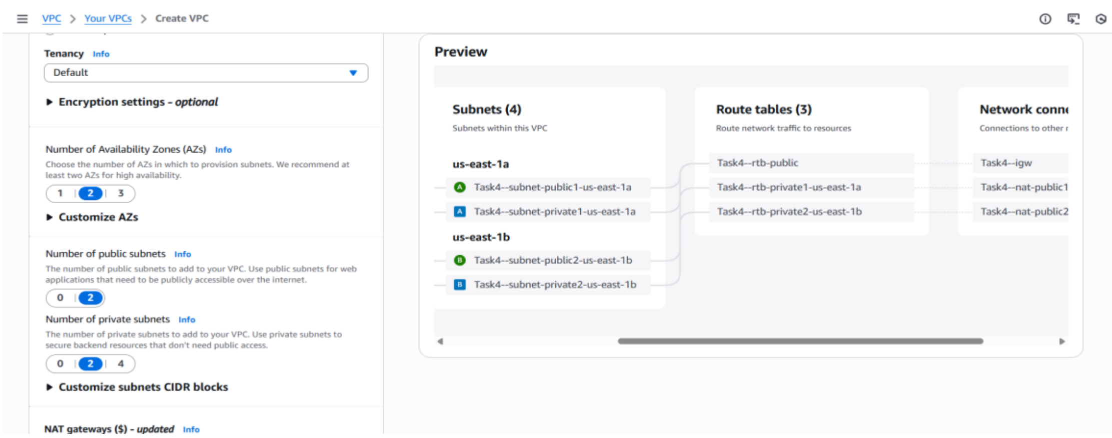
2. Create a custom VPC:
    - Goto – VPC- Create VPC- Select VPC and More
    - Name – my-vpc

      Assign IPV4 CIDR Block – 10.0.0.0/16

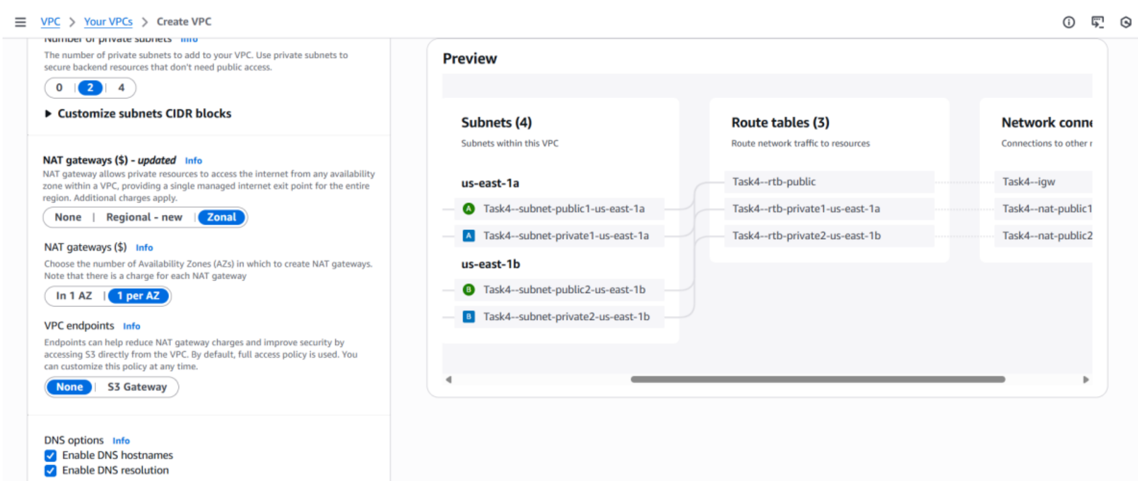
    - (*CIDR block must be in between /16 to /24)*
    - Tendency – Default
    - This will create 2 public subnet, 2 private Subnet, NAT Gateway and an Internet Gateway.



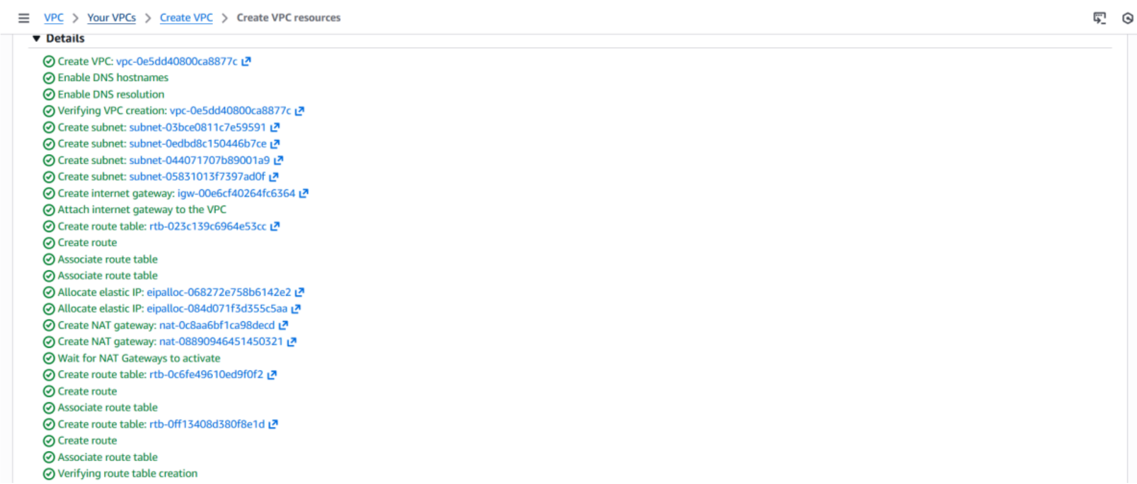
Chose Subnets



Chose NAT Gateway and make sure each AZ have seprate NAT Gateway.



Wait till all process of VPC creation is done



3. Create Security Groups:

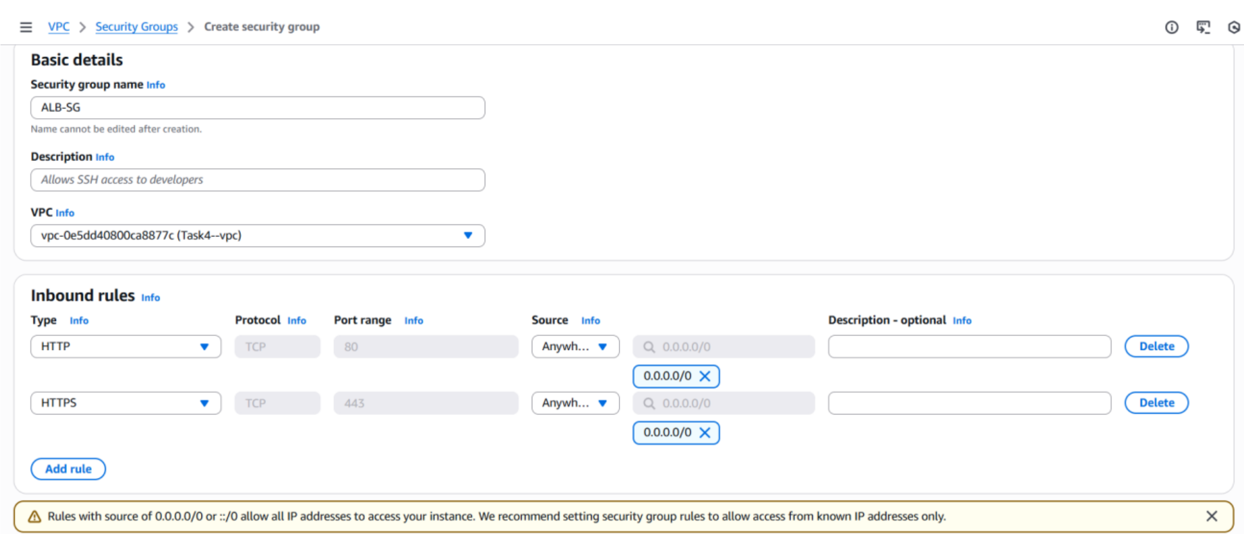
   Application Load Balancer Security Group

   EC2 → Security Groups → Create

   Name- ALB-SG

   Inbound – HTTP-80-0.0.0.0/0

   Outbound – All traffic



   EC2 Security Group

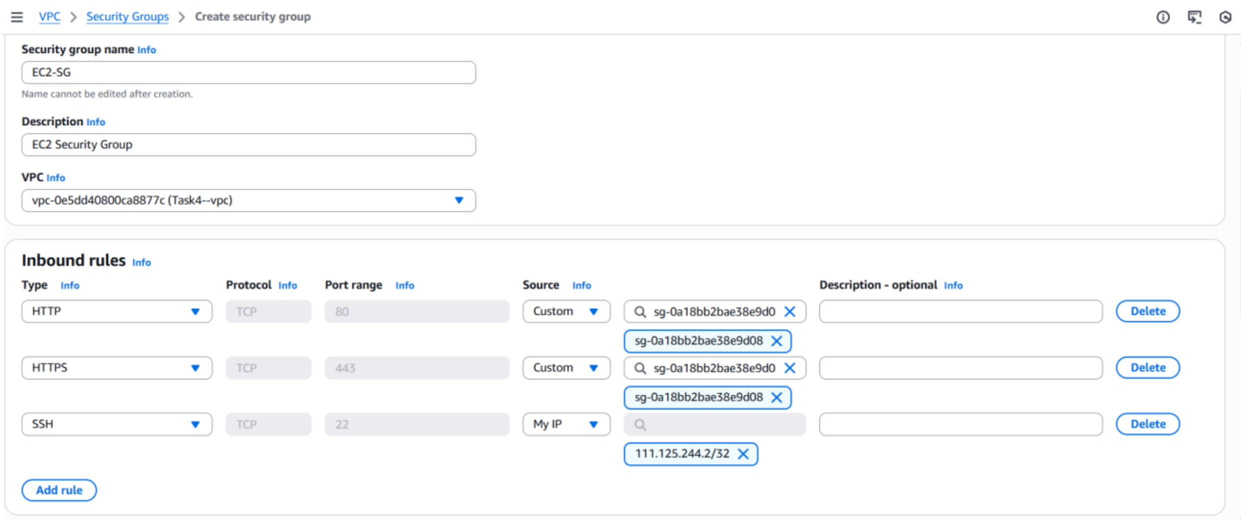
   EC2 → Security Groups → Create

Name- EC2-SG

Inbound – HTTP-80-source: ALB-SG

SSH-22-Source: Your IP

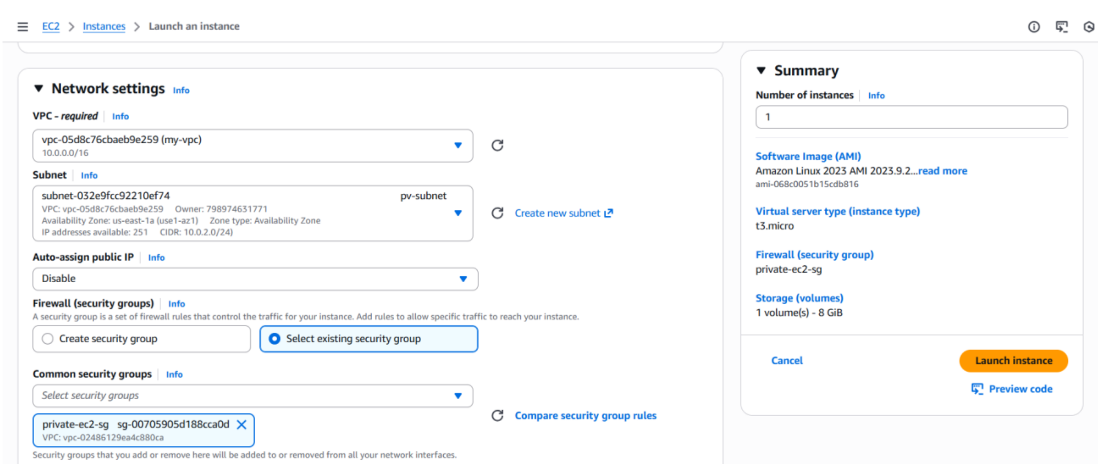
Outbound – All traffic



4. Launch Private EC2 Instance in ec2 private subnet

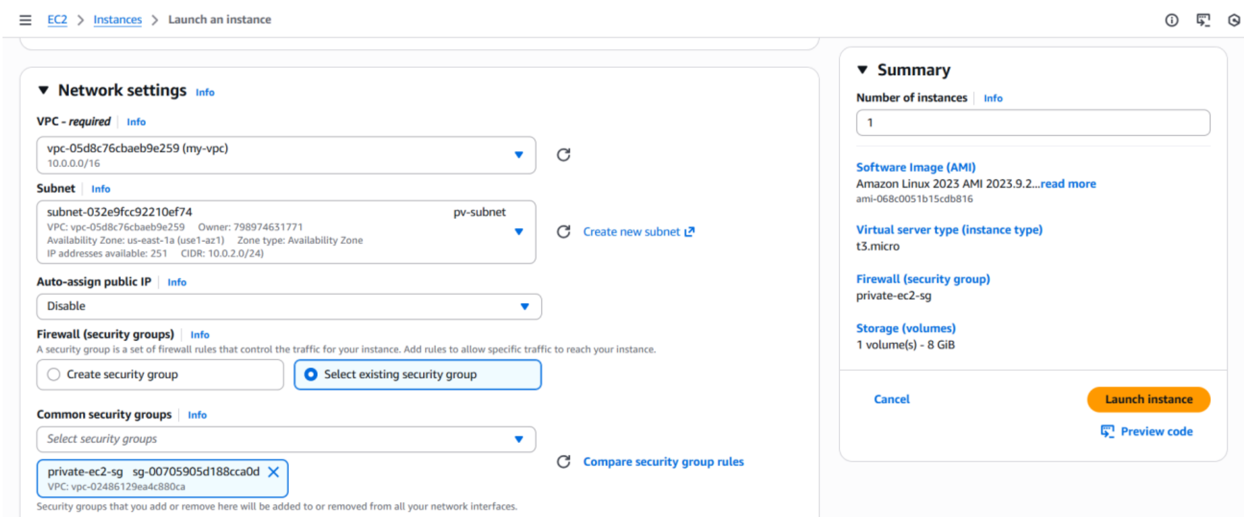
   EC2- Lauch Instance-

   - Name- Privateec2-1
   - OS- Amazon Linux
   - T2-micro
   - Select Key Pair
   - Select Existing Security Group- EC2-SG



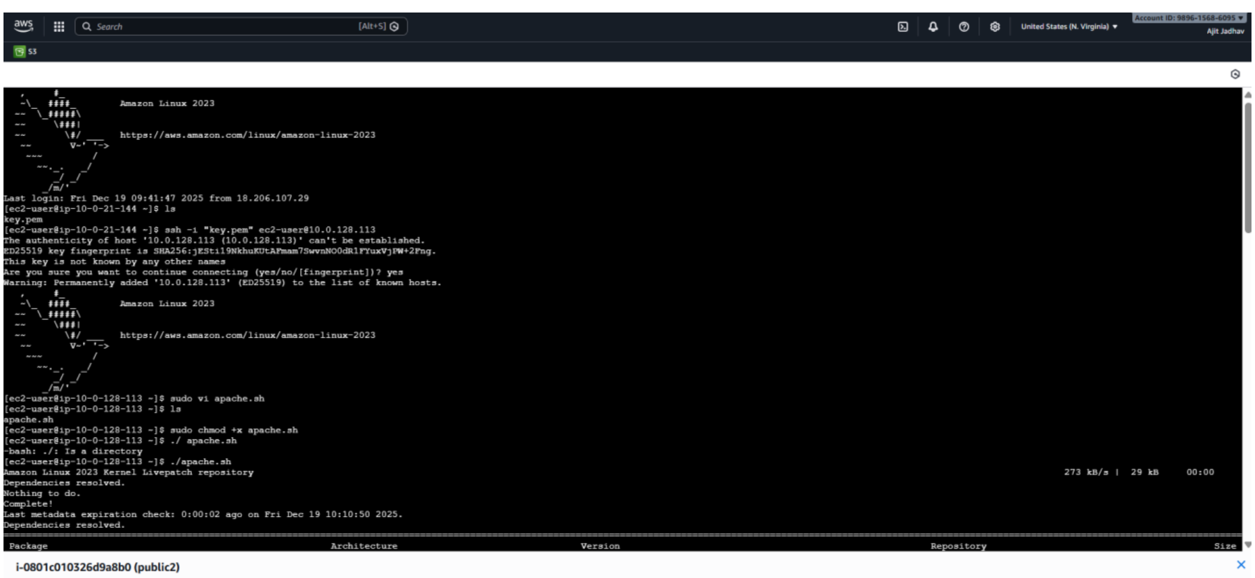
EC2- Lauch Instance-

- Name- Privateec2-2
- OS- Amazon Linux
- T2-micro
- Select Key Pair
- Select Existing Security Group- EC2-SG

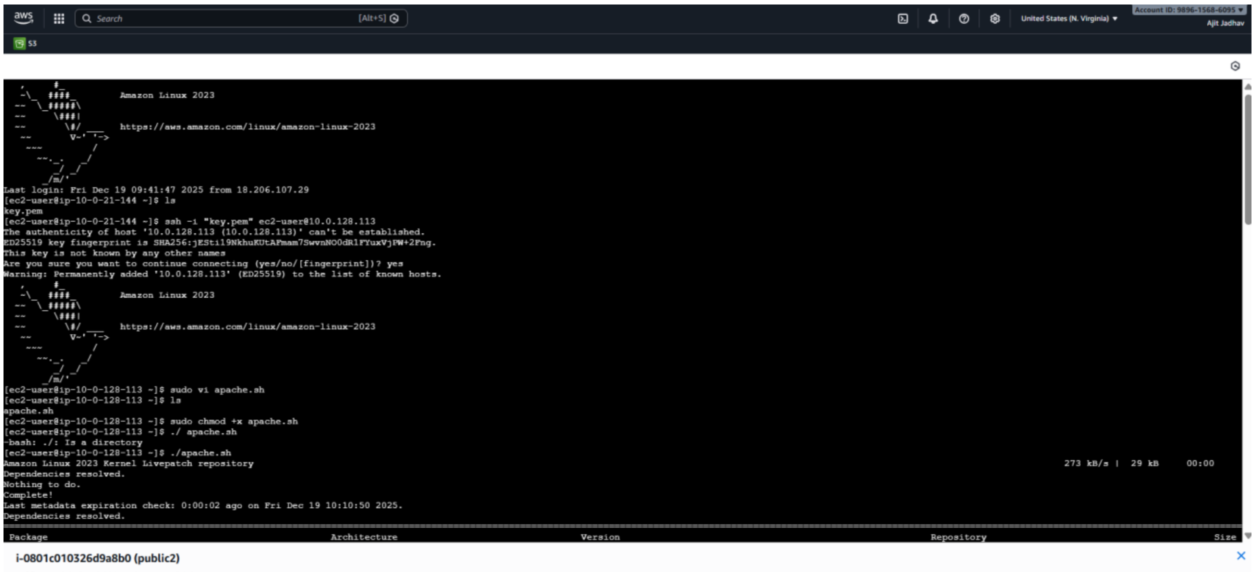


5. Connect These instances using public server

Private server 1

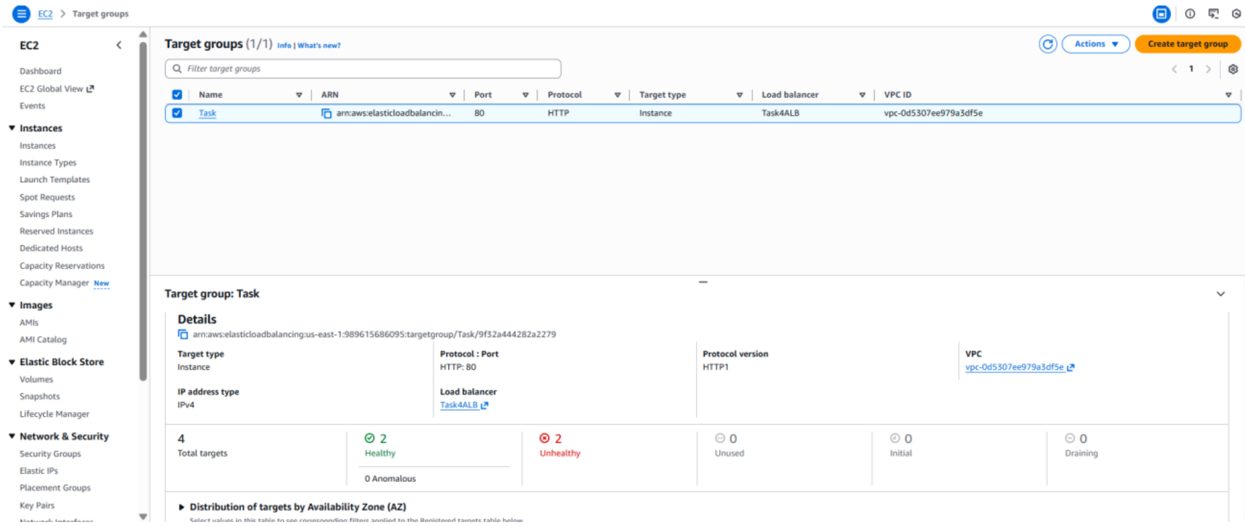


Private server 2

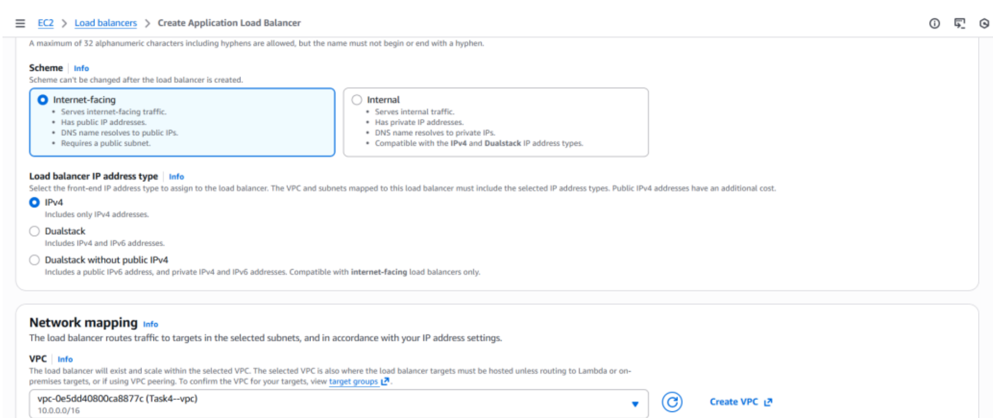


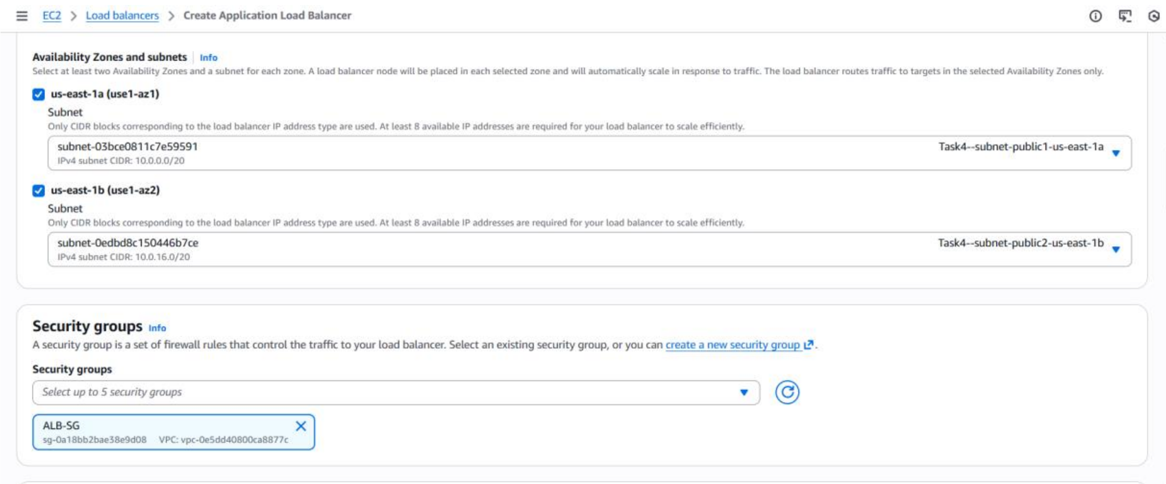
Make sure you install Apache server on both private instance

6. Create target groups
    - EC2 → Target Groups → Create
    - Target type: **Instances**
    - Protocol: HTTP
    - Port: 80
    - VPC: Select your VPC
    - Health check path: /
    - Click **Create**



7. Create Application Load Balancer
    - EC2 → Load Balancers → Create
    - Choose **Application Load Balancer**
    - Name: *Task-APB*
    - Scheme: **Internet-facing**
    - IP type: IPv4
    - Listeners: HTTP 80
    - Subnets:
    - Select **2 public subnets (different AZs)**
    - Security group: ALB-SG
    - Listener action:
    - Forward to **Target Group**
    - Create Load Balancer





8. Check your Load balancer working through a websites

Goto Load Balancer

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Select your Application Load Balancer

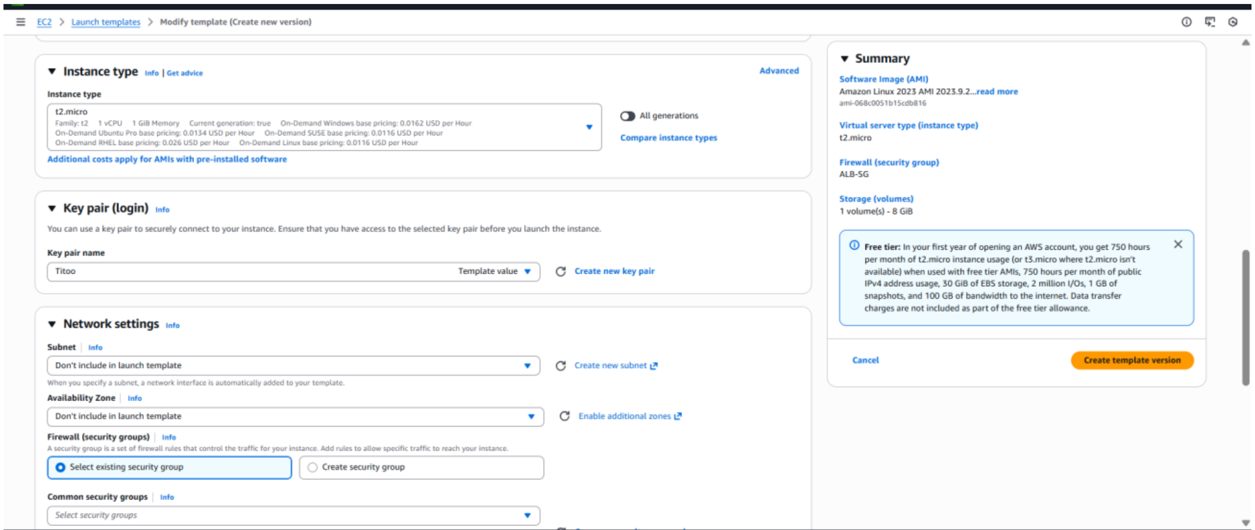
|

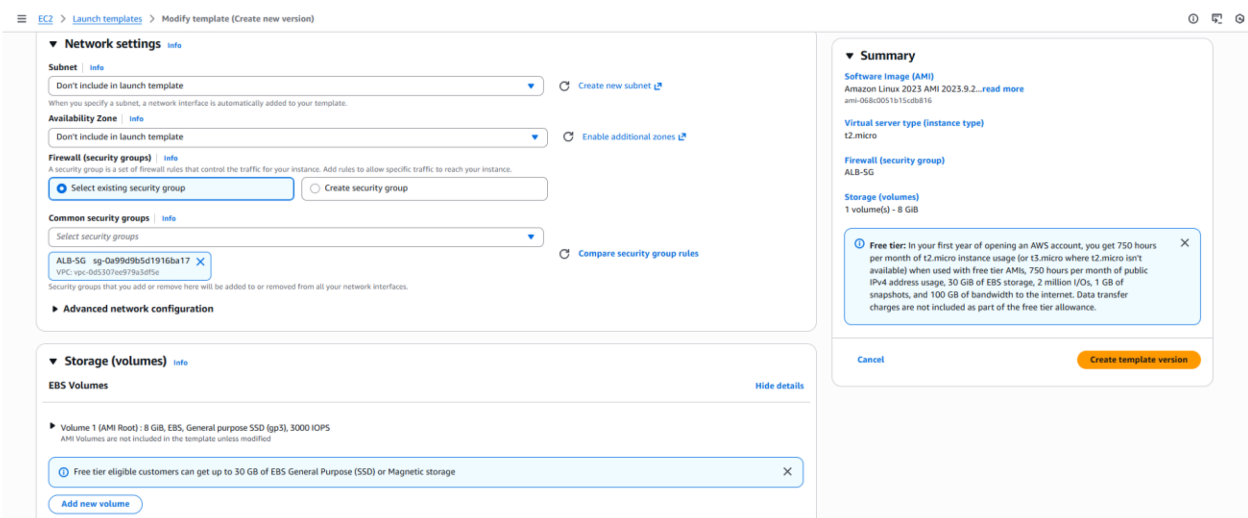
Pick DNS name provided by ALB

paste into web browser (http://)

9. Create a launch Template helps to launch server

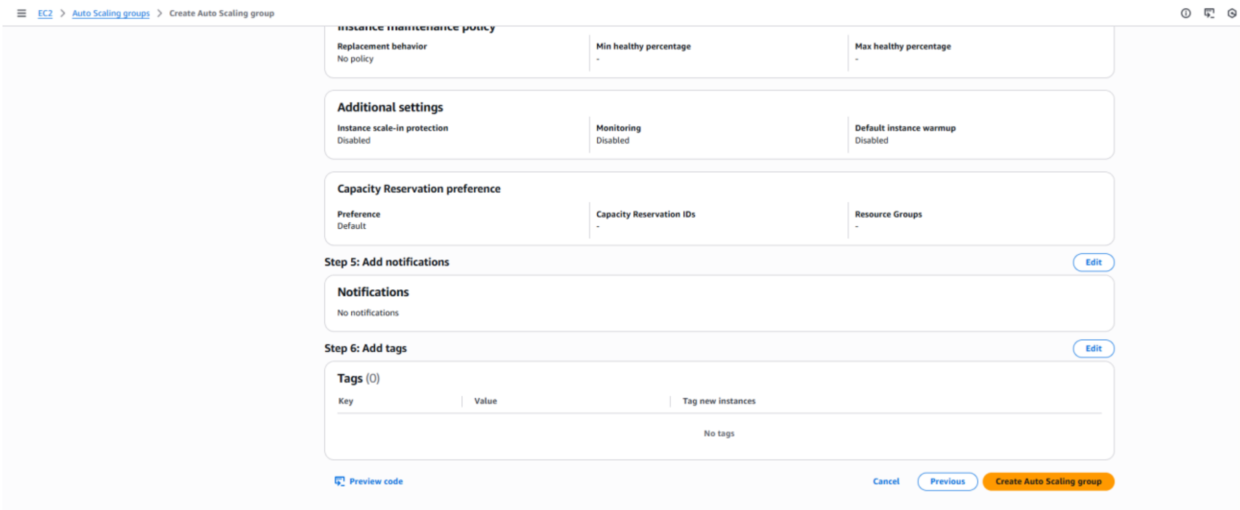
Goto – EC2- Launch Template – Create Template

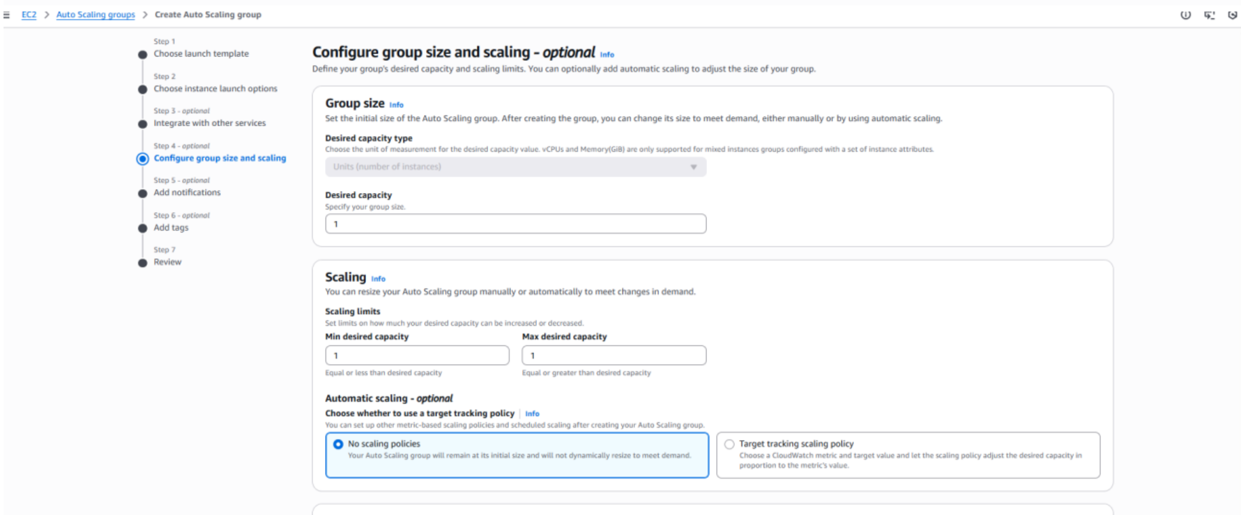


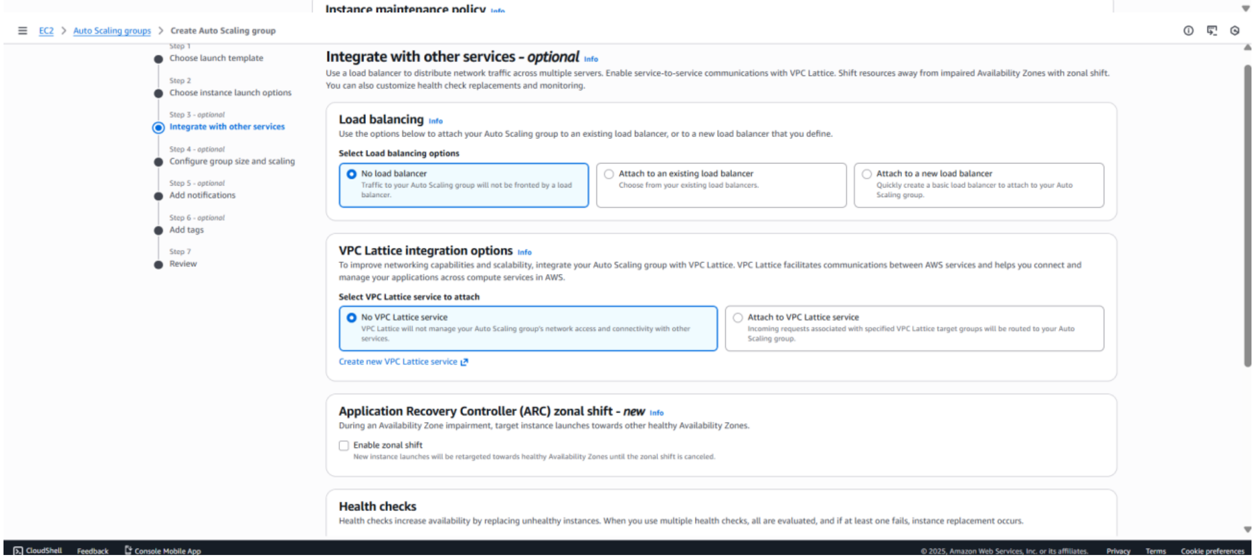


10. Create Auto Scaling Group

EC2- Auto Scaling groups- Create


EC2 > Auto Scaling groups > Create Auto Scaling group
Instance maintenance policy
Replacement behavior
No policy
Min healthy percentage
-
Max healthy percentage
-
Additional settings
Instance scale-in protection
Disabled
Monitoring
Disabled
Default instance warmup
Disabled
Capacity Reservation preference
Preference
Default
Capacity Reservation IDs
-
Resource Groups
-
Step 5: Add notifications
Edit
Notifications
No notifications
Step 6: Add tags
Edit
Tags (0)
Key
Value
Tag new instances
No tags
Preview code
Cancel
Previous
Create Auto Scaling group


EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Configure group size and scaling - optional Info
Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.
Group size Info
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.
Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.
Units (number of instances)
Desired capacity
Specify your group size.
1
Scaling Info
You can resize your Auto Scaling group manually or automatically to meet changes in demand.
Scaling limits
Set limits on how much your desired capacity can be increased or decreased.
Min desired capacity
1
Equal or less than desired capacity
Max desired capacity
1
Equal or greater than desired capacity
Automatic scaling - optional
Choose whether to use a target tracking policy Info
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.
No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.
Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.
Instance maintenance policy Info


EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Integrate with other services - optional Info
Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.
Load balancing Info
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.
Select Load balancing options
No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.
Attach to an existing load balancer
Choose from your existing load balancers.
Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.
VPC Lattice integration options Info
To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.
Select VPC Lattice service to attach
No VPC Lattice service
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.
Attach to VPC Lattice service
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.
Create new VPC Lattice service
Application Recovery Controller (ARC) zonal shift - new Info
During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.
Enable zonal shift
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.
Health checks
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
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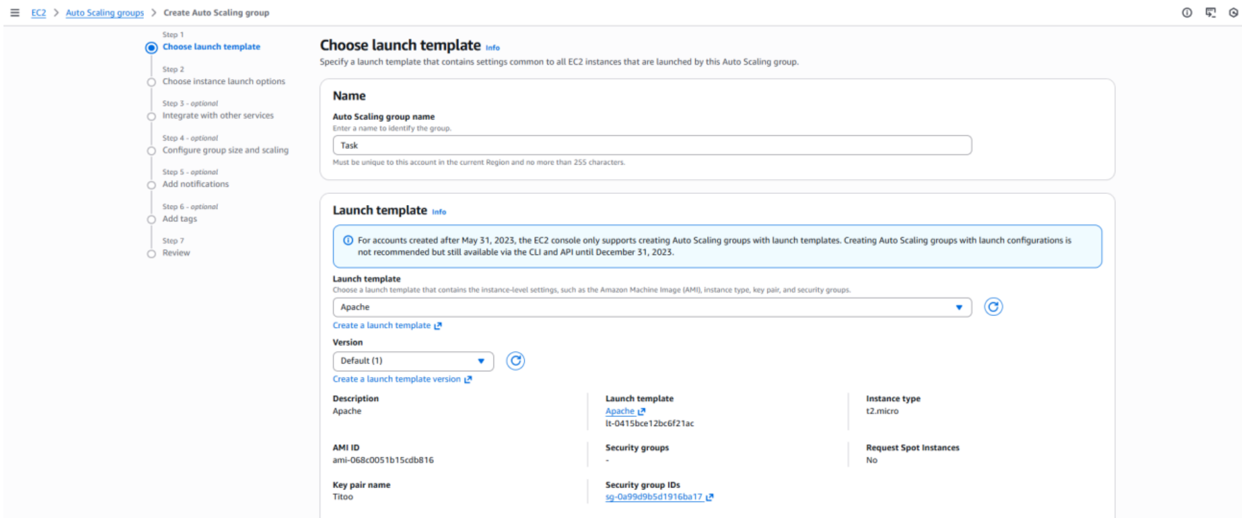
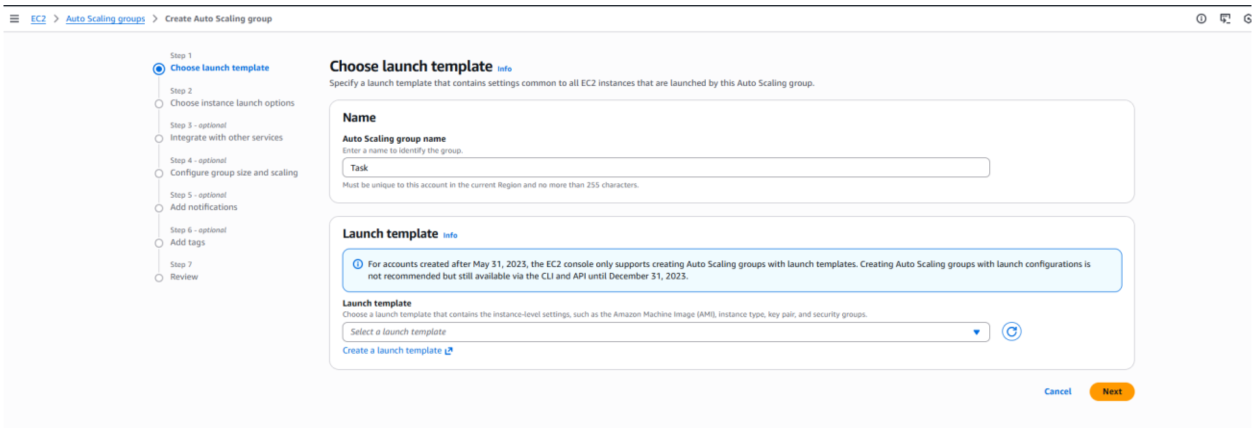




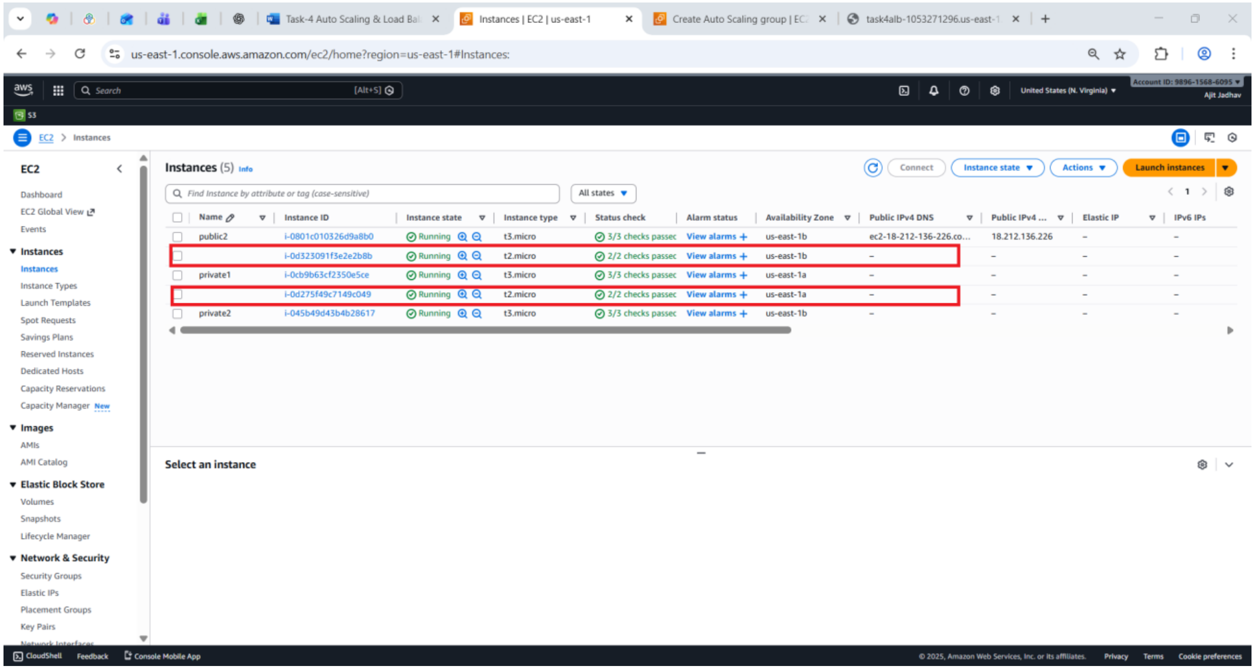
For Checking All Setup

Increase the Stress on instance

sudo apt install stress-ng

stress-ng --cpu 0 --cpu-method all --metrics-brief

Check the Server , New Server will launch as per Existing server crash



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